Designing a Net-Zero Electricity Future for Los Angeles: A Technical and Economic Feasibility Study

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1. Introduction

Los Angeles (LA) is the second-largest city in the United States, with an estimated population of approximately 3.8 million in 2023. It anchors a sprawling metropolitan area that totals about 12.8 million residents. Reflecting its vast size, the gross domestic product (GDP) of the Los Angeles–Long Beach–Anaheim metropolitan region stood at roughly $1.3 trillion in 2023, making it one of the world's largest urban economies (U.S. Bureau of Economic Analysis, 2023).

One distinctive feature of LA's energy landscape is that it owns and operates its own municipal utility, the Los Angeles Department of Water and Power (LADWP)—the largest municipal utility in the nation. LADWP maintains about 8,100 MW of electric generation capacity and delivers on the order of 26 million MWh of electricity per year to roughly 1.4 - 1.5 million customer accounts (serving over four million residents and businesses) as noted by LADWP. This electricity powers LA's vast building stock, infrastructure, and industries, making the power sector a primary focus for local efforts to reduce carbon emissions and transition to cleaner energy.

Historically, LA relied heavily on fossil-fuel-based electricity generation. However, the city has made notable progress in reducing its carbon footprint. According to a recent greenhouse gas (GHG) inventory, community-wide emissions in LA stood at about 26.2 million metric tons of CO_2-equivalent (MMT CO_2e) in 2022 —a 30% decrease from 1990 levels (City of Los Angeles Bureau of Sanitation, 2024). Stationary energy (including buildings and electricity generation) accounted for around 15.6 MMT CO_2e in 2022, though its emissions have dropped by approximately 40% relative to 1990, largely due to the retirement of coal-fired capacity and the increased use of renewable energy in LADWP's power mix. Between 2014 and 2022, the carbon intensity (CO_2 per MWh) of LA's electricity declined by nearly 47%, reflecting the push toward decarbonization. By 2023, LADWP estimated that 45% of its power came from renewable sources—up from 33% in 2020—in line with California's renewable energy mandates (KALALE, 2023)

Beyond the climate rationale, LA has additional reasons to continue ramping up its clean energy transition. The metropolitan region has long struggled with air quality challenges (notably ozone smog and particulate matter), and fossil-fuel combustion in the electricity sector and transportation sector remains a major contributor to poor air quality. Efforts to replace coal- and natural gas-fired plants with wind, solar, and other low-carbon resources can help reduce these harmful pollutants. As a result, the shift to clean power brings public health benefits, including lower rates of asthma and respiratory diseases.

The goal of this project is to determine how Los Angeles can achieve a net-zero energy system for its electricity needs by increasing renewable generation capacity—focusing specifically on solar and wind resources. By examining LADWP's current load profile and estimating how much of the existing demand is met by fossil fuels, we can then propose a portfolio of wind and solar projects that fully displace non-renewable generation over the course of a year. The following sections detail our methodology for analyzing LA's energy demand, solar and wind potential, and the resulting pathways to supply 100% of the city's electricity from local renewable energy sources.

Method

This study employed a four-step approach to determine how Los Angeles can achieve a net-zero electricity system using solar and wind. First, LA's monthly and hourly demand was analyzed using EIA datasets to identify patterns, peak loads, and any potential deficits not covered by current renewables. Second, solar and wind potential were evaluated, drawing on publicly available datasets (NSRDB for solar and NREL's Offshore Toolkit for wind) and using established tools (e.g., Weibull distribution for wind speed and irradiance-based calculations for solar). Third, these resource profiles were compared with the demand timeline to calculate the capacity of solar panels and wind turbines needed to eliminate the gap between demand and existing renewable generation—focusing especially on the worst-case month. Finally, an economic feasibility assessment was conducted using the P1–P2 method. This end-to-end process—demand analysis, resource evaluation, capacity sizing, and cost modeling—provides a robust framework for projecting the technical and economic viability of scaling solar and wind resources in Los Angeles.

Energy Demand

The U.S. Energy Information Administration (EIA) publishes LADWP's operational data through its Open Data platform, allowing us to access both region-level and fuel-type datasets.

- Region-Level Data includes:
    - Total Demand (D): Electricity consumed within LADWP's territory.
    - Forecast Demand (DF): Day-ahead demand projections.
    - Net Generation (NG): Total electricity produced by local generators.
    - Interchange (TI): Net power flows imported (positive) or exported (negative).
- Fuel-Type Data indicates how much electricity is generated from each primary resource:
    - Coal (COL)
    - Natural Gas (NG)
    - Solar (SUN)
    - Wind (WND)
    - Hydro (WAT)
    - Other (OTH)

Using the 2023–2024 LADWP daily generation data, a daily generation by fuel type plot was created as shown in Figure 1, illustrating how coal, natural gas, solar, wind, hydro, and other sources contribute to the total generation each day. Figure 2 presents LADWP's daily demand, forecast, generation, and interchange, providing an overview of how supply meets demand on a day-to-day basis. Figure 3 goes a step further by illustrating hourly demand, forecast, generation, and interchange patterns, thereby revealing short-term fluctuations within a single day. However, as shown in Figures 2 and 3, from June 28 to July 24 the daily demand and net generation both remain at or near 4,200 MWh, while interchange is essentially zero. In addition, Figure 1 indicates a total absence of fuel-type data during the same window. These observations suggest either missing data or placeholder values returned by the API, rather than a genuine reflection of LADWP's actual operational conditions.

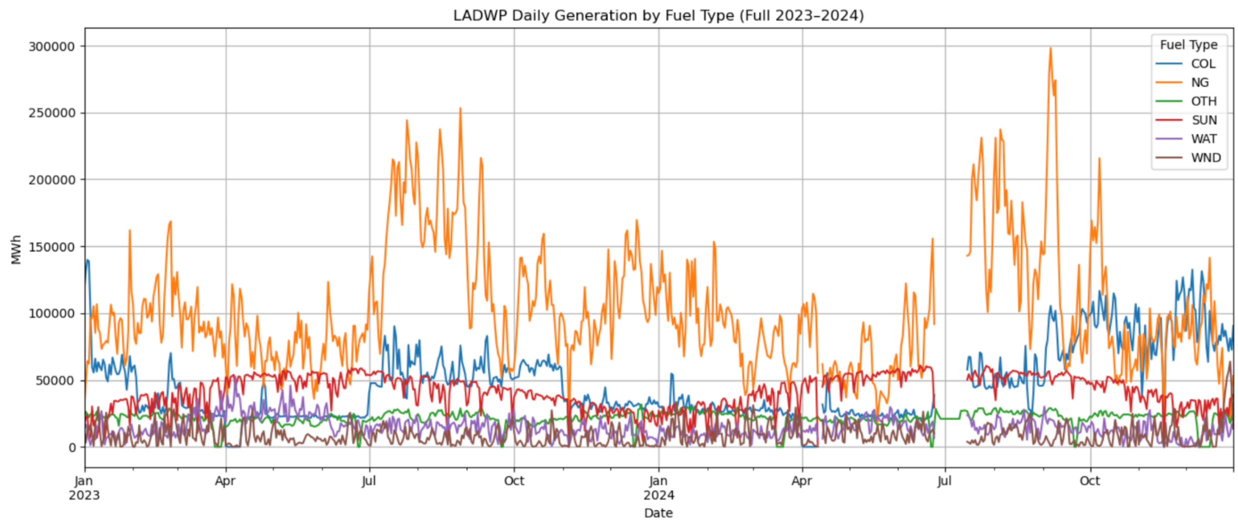

Figure 1: LADWP daily generation by fuel type

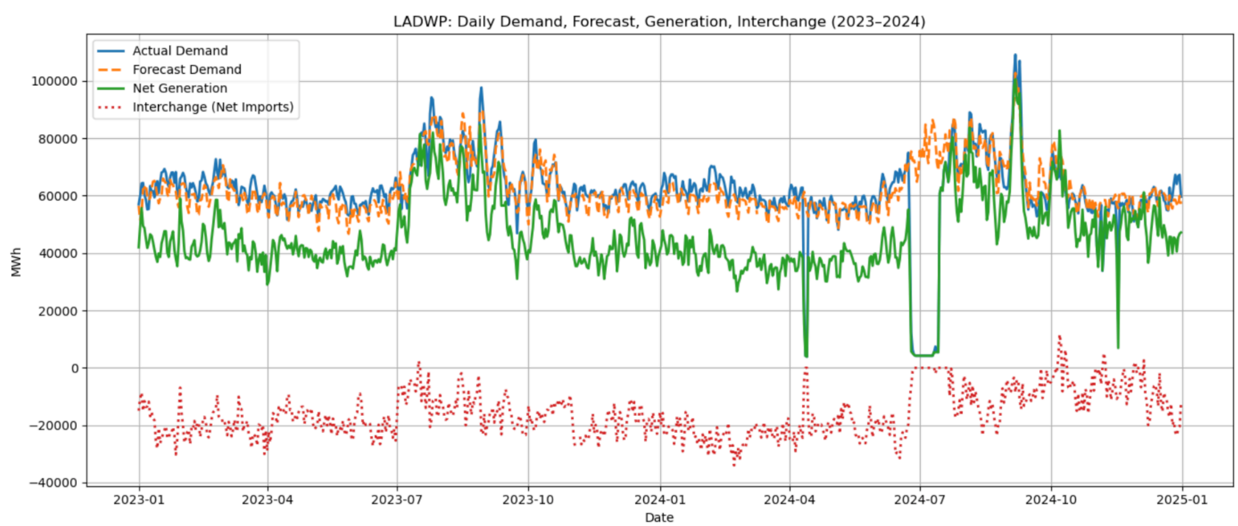

Figure 2: LADWP daily demand, forecast, generation, and interchange

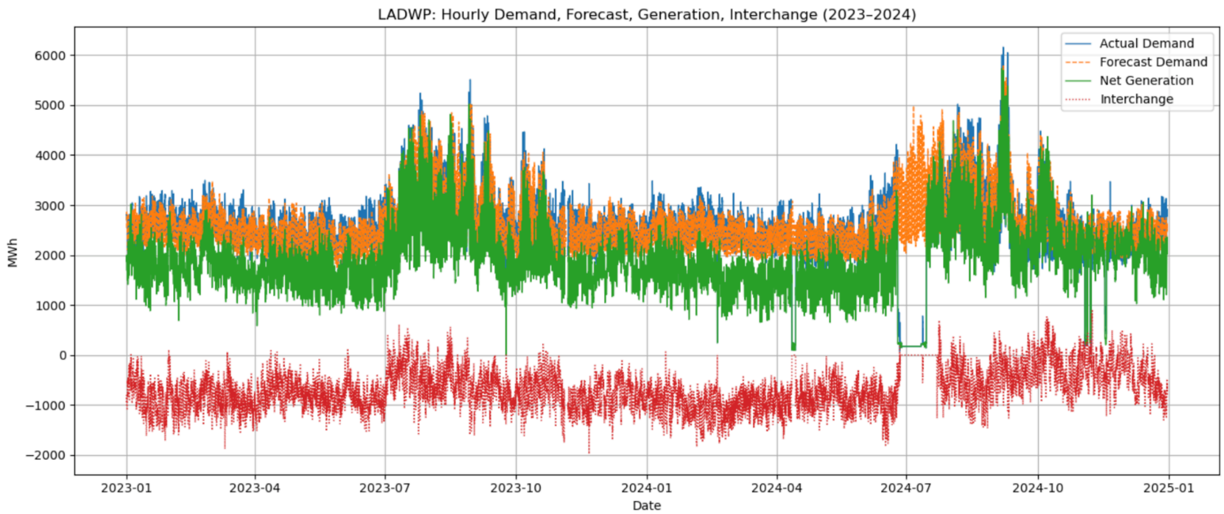

Figure 3: LADWP hourly demand, forecast, generation, and interchange

After obtaining hourly region and fuel-type data for the 2023–2024 period, we combined them on a common "Date" column to produce a single dataset. This unified view shows, for each hour, LADWP's total demand, total net generation, and the specific contributions from coal, natural gas, solar, wind, hydro, and other sources. Analyzing both region-level and fuel-type metrics side by side helps us identify whether LADWP's supply mix meets the city's real-time energy needs.

Within this merged dataset, we created several new columns to characterize different supply–demand relationships:

- diff_demand_minus_netgen: The gap between total demand (D) and net generation (NG), indicating whether total local generation is sufficient to meet demand hour by hour.
- non_renewable: The total electricity generated by coal (COL), natural gas (NatGas), and other miscellaneous fuels (OTH).
- diff_demand_minus_nonrenewable: The portion of demand not met if only these non-renewable sources are used.
- renewable: The combined sum of solar (SUN), wind (WND), and hydro (WAT).
- diff_demand_minus_renewable: The amount of additional renewable energy needed—i.e., how much demand remains uncovered after accounting for all solar, wind, and hydro output.

For this project, we are not considering interchange (TI), because our main focus is on developing a system in which local renewable energy sources alone can ultimately supply the current non-renewable portion of the load. An example snippet of this merged data is shown in Table 1.

Table 1: Example Rows from the Combined Hourly LADWP Dataset

Date	D	COL	NatGas	OTH	SUN	WAT	WND	diff_demand_minus_netgen	non_renewable	diff_demand_minus_nonrenewable	renewable	diff_demand_minus_renewable
2023-01-01 00:00:00	2712.0	916.0	250.0	212.0	66.0	0.0	255.0	1013.0	1378.0	1334.0	321.0	2391.0
2023-01-01 01:00:00	2813.0	1051.0	392.0	220.0	10.0	0.0	243.0	897.0	1663.0	1150.0	253.0	2560.0
2023-01-01 02:00:00	2801.0	1069.0	474.0	223.0	0.0	0.0	234.0	801.0	1766.0	1035.0	234.0	2567.0
2023-01-01 03:00:00	2824.0	1066.0	381.0	222.0	0.0	0.0	196.0	959.0	1669.0	1155.0	196.0	2628.0
2023-01-01 04:00:00	2782.0	1144.0	381.0	219.0	0.0	0.0	239.0	799.0	1744.0	1038.0	239.0	2543.0
2023-01-01 05:00:00	2803.0	1084.0	328.0	209.0	0.0	0.0	249.0	933.0	1621.0	1182.0	249.0	2554.0
2023-01-01 06:00:00	2731.0	953.0	210.0	210.0	0.0	0.0	256.0	1102.0	1373.0	1358.0	256.0	2475.0
2023-01-01 07:00:00	2638.0	907.0	224.0	220.0	0.0	0.0	250.0	1037.0	1351.0	1287.0	250.0	2388.0
2023-01-01 08:00:00	2540.0	873.0	239.0	216.0	0.0	0.0	245.0	967.0	1328.0	1212.0	245.0	2295.0
2023-01-01 09:00:00	2433.0	941.0	214.0	187.0	0.0	0.0	210.0	881.0	1342.0	1091.0	210.0	2223.0
2023-01-01 10:00:00	2353.0	965.0	207.0	156.0	0.0	3.0	199.0	823.0	1328.0	1025.0	202.0	2151.0
2023-01-01 11:00:00	2269.0	953.0	188.0	156.0	0.0	228.0	215.0	529.0	1297.0	972.0	443.0	1826.0

While analyzing at the hourly level reveals short-term fluctuations and operational insights, aggregating these data into daily or monthly totals highlights broader, long-term trends. For instance, by summing across each day, we can see how demand compares to generation on a day-to-day basis and observe typical patterns or anomalies. Monthly summaries, as shown in Table 2, help illuminate seasonal variations, such as changes in solar irradiance or wind speeds, which could create sustained energy shortfalls or surpluses. Adopting this tiered approach—hourly, daily, and monthly—gives both the detail needed to understand short-term dynamics and the context for long-term planning.

Table 2: Combined Monthly LADWP Dataset

Date	D	COL	NatGas	OTH	SUN	WAT	WND	diff_demand_minus_netgen	non_renewable	diff_demand_minus_nonrenewable	renewable	diff_demand_minus_renewable
2023-01-31	1963949.0	432860.0	524552.0	143278.0	148037.0	61200.0	62102.0	591920.0	1100690.0	863259.0	271339.0	1692610.0
2023-02-28	1806346.0	210541.0	613248.0	132492.0	184941.0	59406.0	49454.0	556264.0	956281.0	850065.0	293801.0	1512545.0
2023-03-31	1937027.0	179695.0	571257.0	111080.0	231396.0	124008.0	68924.0	650667.0	862032.0	1074995.0	424328.0	1512699.0
2023-04-30	1726237.0	102747.0	479265.0	119081.0	310892.0	171192.0	40845.0	502215.0	701093.0	1025144.0	522929.0	1203308.0
2023-05-31	1810039.0	141046.0	405117.0	110825.0	320495.0	169138.0	44737.0	618681.0	656988.0	1153051.0	534370.0	1275669.0
2023-06-30	1769641.0	135230.0	464771.0	111174.0	310302.0	98019.0	46648.0	603497.0	711175.0	1058466.0	454969.0	1314672.0
2023-07-31	2281597.0	357554.0	1031090.0	147250.0	313452.0	97324.0	53852.0	281075.0	1535894.0	745703.0	464628.0	1816969.0
2023-08-31	2406354.0	355097.0	1116593.0	144633.0	272308.0	95593.0	33114.0	389016.0	1616323.0	790031.0	401015.0	2005339.0
2023-09-30	2024685.0	332991.0	670038.0	123835.0	248290.0	83213.0	48913.0	517405.0	1126864.0	897821.0	380416.0	1644269.0
2023-10-31	1998798.0	369682.0	692806.0	117522.0	239150.0	77492.0	36233.0	465913.0	1180010.0	818788.0	352875.0	1645923.0
2023-11-30	1681367.0	185017.0	504586.0	125314.0	165284.0	66986.0	27769.0	606411.0	814917.0	866450.0	260039.0	1421328.0
2023-12-31	1890301.0	193315.0	761710.0	136883.0	142667.0	60583.0	25595.0	569548.0	1091908.0	798393.0	228845.0	1661456.0
2024-01-31	1938048.0	192059.0	644956.0	162354.0	155363.0	60819.0	51408.0	671089.0	999369.0	938679.0	267590.0	1670458.0
2024-02-29	1850296.0	161127.0	508517.0	139880.0	155982.0	61958.0	60121.0	762711.0	809032.0	1040772.0	278061.0	1571743.0

2024-03-31	1805971.0	162942.0	449569.0	108086.0	242345.0	63982.0	82797.0	696250.0	720597.0	1085374.0	389124.0	1416847.0
2024-04-30	1599127.0	98928.0	391252.0	121520.0	267370.0	72776.0	44567.0	602714.0	600912.0	987427.0	384713.0	1203626.0
2024-05-31	1732957.0	145553.0	344177.0	119860.0	333578.0	126348.0	59559.0	603882.0	609590.0	1123367.0	519485.0	1213472.0
2024-06-30	1587946.0	117816.0	428367.0	117365.0	267670.0	68617.0	52616.0	535495.0	634408.0	911290.0	388903.0	1156795.0
2024-07-31	1272884.0	177931.0	572460.0	149600.0	184779.0	51537.0	23893.0	112684.0	836844.0	370867.0	260209.0	947502.0
2024-08-31	2346342.0	302233.0	975387.0	159159.0	331088.0	100942.0	67242.0	410291.0	1436779.0	909563.0	499272.0	1847070.0
2024-09-30	2089726.0	481477.0	806788.0	125474.0	312082.0	95759.0	35226.0	232936.0	1413739.0	675987.0	443067.0	1646659.0
2024-10-31	1915054.0	583283.0	615195.0	148385.0	284716.0	69813.0	45708.0	167954.0	1346863.0	568191.0	400237.0	1514817.0
2024-11-30	1561089.0	498541.0	480881.0	142330.0	200967.0	52040.0	52242.0	134088.0	1121752.0	439337.0	302349.0	1247612.0
2024-12-31	1794435.0	555100.0	469308.0	101001.0	180979.0	50840.0	85527.0	351680.0	1125409.0	669026.0	317346.0	1477089.0

Having identified the “renewable energy needed” via diff_demand_minus_renewable, we propose two primary options for bridging the gap currently covered by non-renewables: solar and wind. In the following chapters, we will explore where and how LADWP might expand solar and wind capacity to move toward a clean, self-sufficient energy mix.

Wind

Data Acquisition

The wind energy analysis process begins with data acquisition from the National Renewable Energy Laboratory (NREL) Offshore California Wind Toolkit. The data is accessed through an API endpoint: ***https://developer.nrel.gov/api/wind-toolkit/v2/wind/offshore-ca-download.csv.*** We applied specific parameters including an API key, WKT (Well-Known Text) geometry defining the spatial location, a valid email address, and the year for which the data is requested.

Details of the input	
WKT (Coordinates)	**(-121.5 34.12); Offshore LA, CA**
Year	**2019 (Data post 2019 isn't available)**
Parameters	**Windspeed (80m, 100m, 120m), Wind direction**

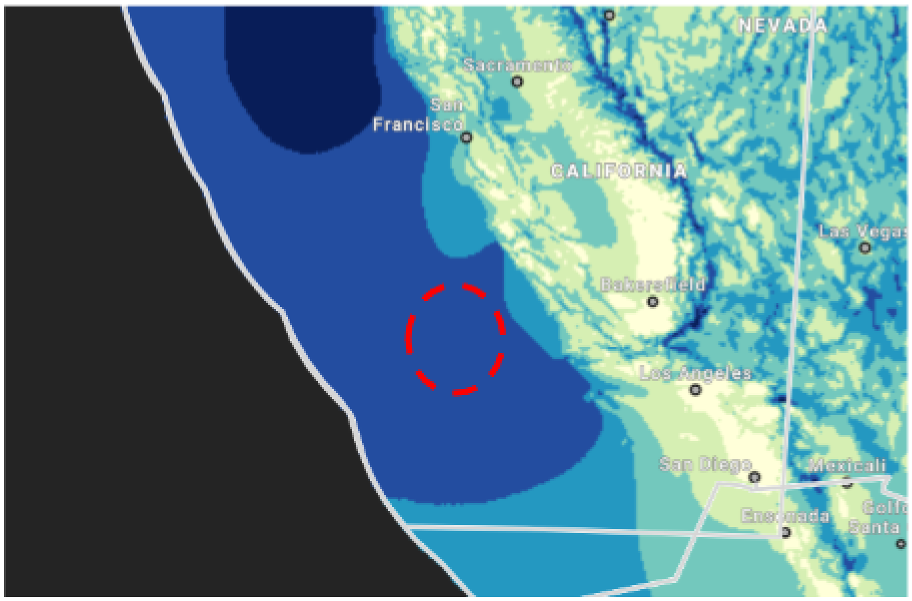

Figure 4: Selection of site for offshore wind turbines (it's off the coast of LA on a bunch of islands)

The dataset provided includes wind speed measurements at three different heights—80 meters, 100 meters, and 120 meters—as well as the corresponding wind direction values, making it well-suited for assessing wind resource potential across various elevations.

Wind Speed Height Normalization

To compare turbine performance across different hub heights, wind speeds were interpolated to a common reference height (140 meters in this case) using the Power Law:

$$V(h) = V(h_0)\left(\frac{h}{h_0}\right)^{\alpha}$$

Where:

- V(h) = wind speed at height h
- V(h0)= wind speed at reference height h
- α = shear exponent (typically 0.14 for offshore, 0.2–0.3 for land)

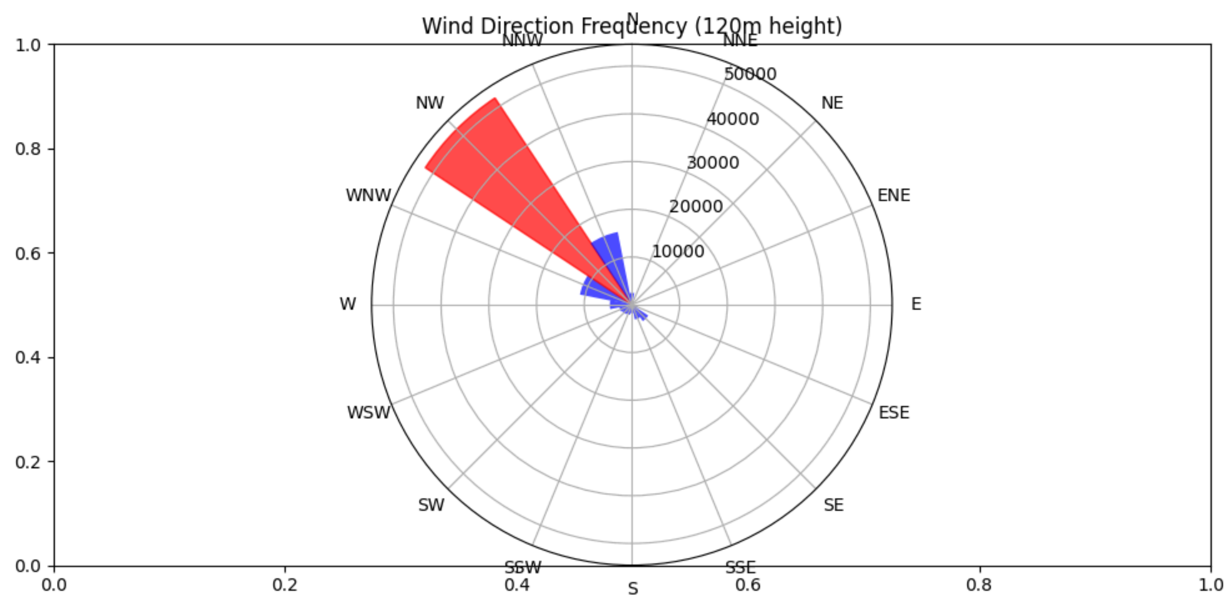

Figure 5: Predominant Wind direction; comes out to be NorthWest

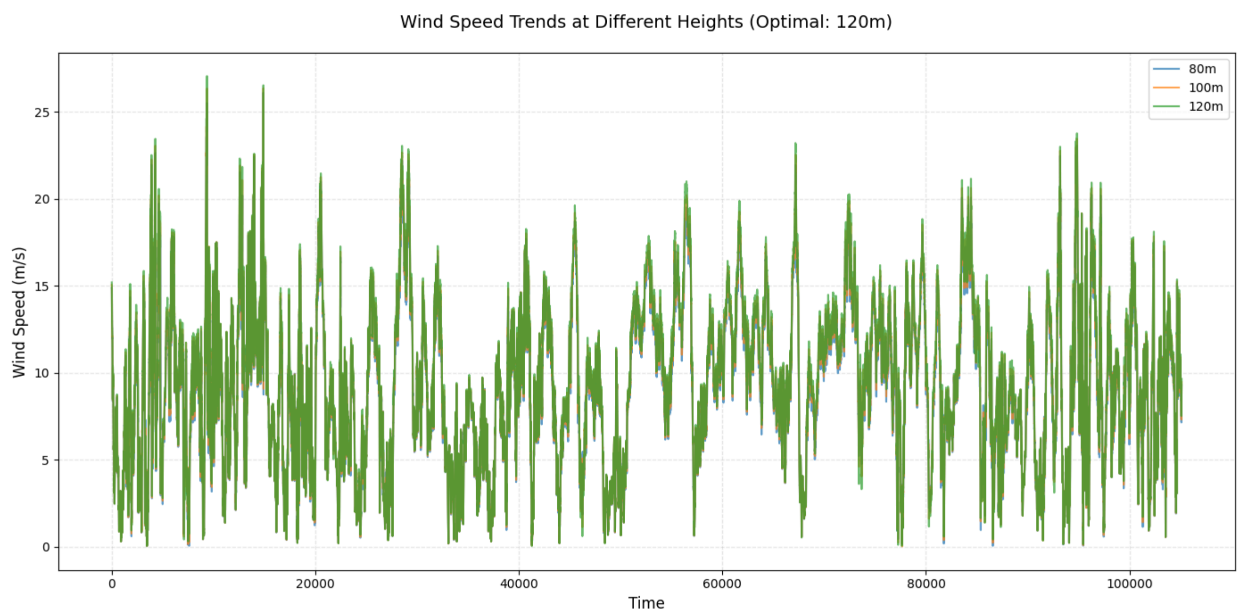

Figure 6: Wind speeds at different levels, optimal wind speed average found at 120m.

Wind Speed Filtering and Outlier Removal

To remove spurious measurements and calm wind periods:

- Zero wind speeds were excluded.
- Outliers were filtered using the interquartile range (IQR) method:

$$\text{Bounds}=[Q1-1.5\times IQR,\ Q3+1.5\times IQR]$$

This step ensures accurate statistical fitting and removes atypical values that could bias the Weibull parameter estimation.

Weibull Distribution Fitting

The Weibull distribution is widely used to describe wind speed variability. It is defined as:

$$f(v;k,c)=\frac{k}{c}\left(\frac{v}{c}\right)^{k-1}\exp\left[-\left(\frac{v}{c}\right)^{k}\right]$$

Where:

- v = wind speed
- k = shape parameter
- c = scale parameter

We used Maximum Likelihood Estimation (MLE) to fit the distribution:

```
params = weibull_min.fit(data, floc=0)

k, loc, c = params
```

Turbine Model and Power Curve Definition

The selected turbine, **Siemens Gamesa SG 14-236 DD**, has the following specifications:

- Rated Power : 14 MW
- Cut-in Speed : 3.5 m/s
- Rated Speed : 13.5 m/s
- Cut-out Speed : 25 m/s

- Rotor Diameter : 236 m
- Hub Height : 132 m

This is one of the common turbines used for offshore wind power generation and has a good capacity.

Figure 7: Siemens Gamesa SG 14-236 DD, one of their biggest

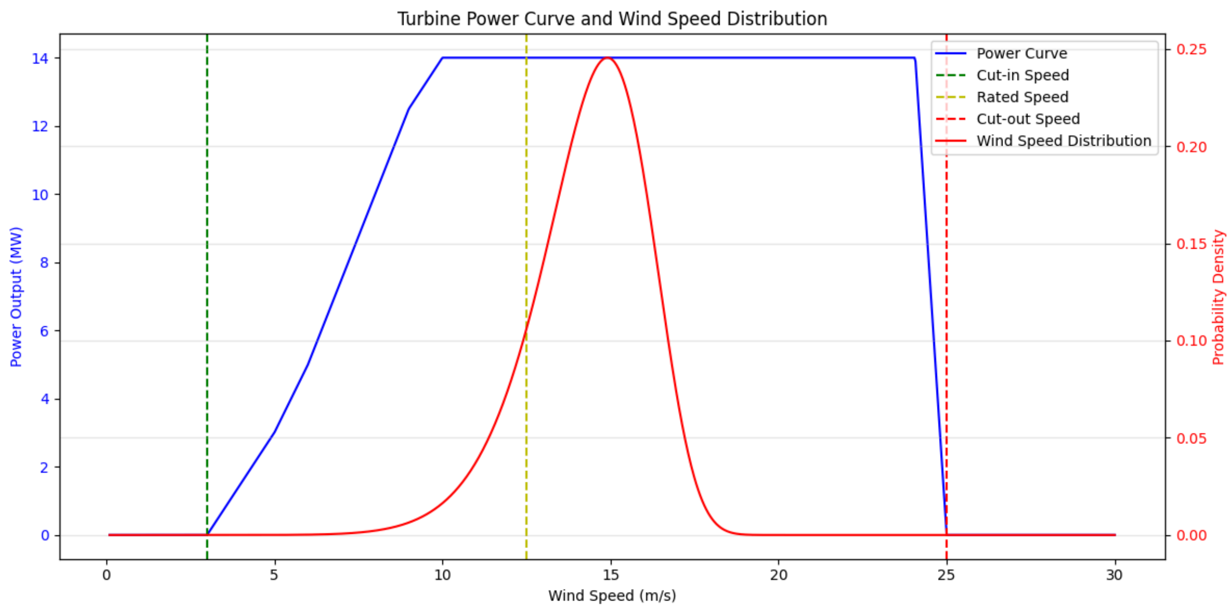

Figure 8: Turbine Power Curve and Wind speed distribution (Probability distribution function)

Energy Production Estimation

To estimate expected annual energy production, we integrated the power output curve weighted by the Weibull probability density function over the full wind speed range:

$$\mathbb{E}[P] = \int_0^{\infty} P(v) \cdot f(v)\, dv$$

We used trapezoidal numerical integration

- expected_power = np.trapezoid(power_values * prob_density, v_range)
- annual_energy_wh = expected_power * 8760

The capacity factor (CF) was calculated as:

$$\text{CF} = \frac{\text{Annual Energy (kWh)}}{P_r \cdot 8760}$$

Monthly Variability and Seasonal Analysis

Monthly Weibull parameters and energy production were computed using the same process, assuming each month has equal duration:

$$E_{month} = \int_0^{\infty} P(v) \cdot f_m(v)\, dv \cdot \left(\frac{8760}{12}\right)$$

With this, bar plots showing monthly energy output and capacity factors were generated.

Table 3: Monthly Wind Statistics and Energy Production for one turbine

Month	**Weibull k**	**Weibull c (m/s)**	**Energy (MWh)**	**Capacity Factor (%)**
January	1.97	9.66	3519.07	34.40%
February	2.46	11.31	4788.98	46.90%
March	2.04	8.81	2885.3	28.20%
April	1.77	10.63	4052.63	39.70%
May	2.35	9.66	3484.65	34.10%
June	1.93	9.24	3231.22	31.60%
July	3.63	13.79	7164.48	70.10%
August	3.11	12.42	5907.57	57.80%
September	5.03	13.33	7540.52	73.80%
October	2.36	12.01	5203.31	50.90%
November	2.08	9.28	3233.25	31.60%
December	2.67	10.82	4465.06	43.70%
		Total	**55476.04**	

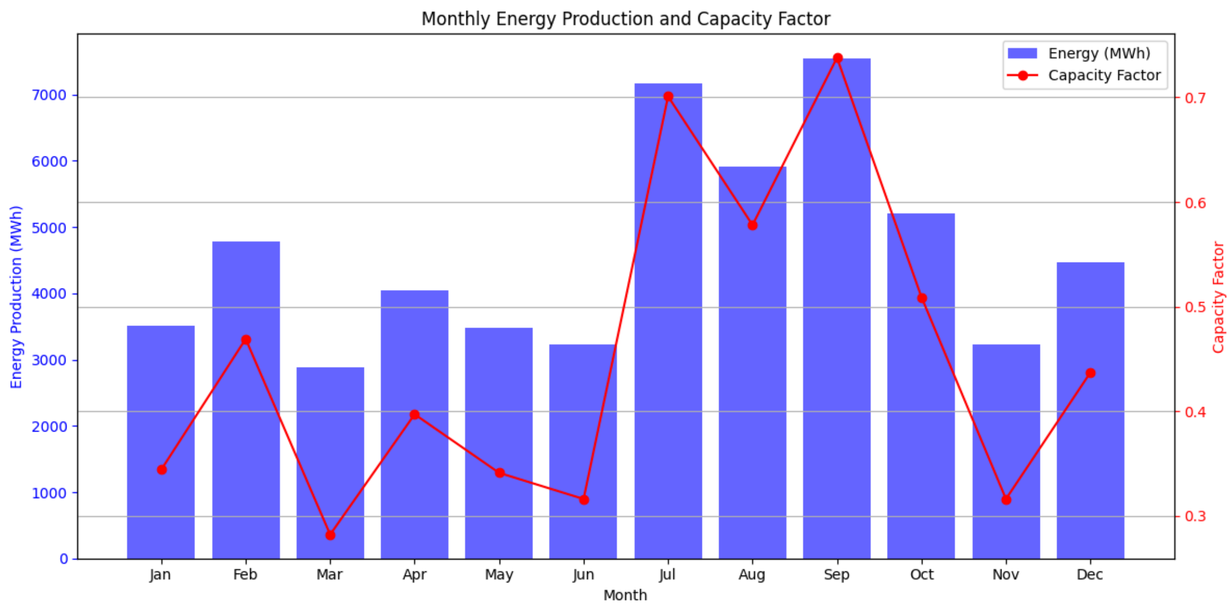

Figure 9: Monthly Energy production and Capacity Factor (CF) for one turbine

Environmental Impact: CO2 Saving Estimation

Assuming a grid emission factor of 0.3 kg CO2 per kWh, we estimated the carbon offset from replacing fossil-fuel-derived electricity:

$$\text{CO}_2\text{ Savings (kg)} = E_{annual\_kWh} \times 0.3$$

Cost

As per PNNL's US distributed wind installed cost.the typical costs of utility-scale wind ranged from $2900/kW. At $2900 per Kilowatt, the total cost of the solar system is about $6.9 billion dollars.

Table 4: Annual Results

Weibull Shape Factor (k)	**2.61**
Weibull Scale Factor (c)	**10.91 m/s**
Annual Energy Output	**53855.90 MWh or 53.85 GWh**
Capacity Factor	**43.91%**

CO_2 Saved	**16156769.31 kg CO_2**

Solar

Data Collection

Solar radiation data was acquired from the National Solar Radiation Database (NSRDB) maintained by NREL. The precise coordinate used was 34.596 for latitude and -117.114 for longitude. We used the 2022 set of data from the GOES Conus: PSM v4 dataset. The hourly data from NSRDB included air temperature, direct normal irradiance, direct horizontal irradiance, global horizontal irradiance, and surface albedo.

Calculations

Using the data from NREL, we calculated additional solar radiation parameters such as the solar hour and the solar zenith angle. For the slope of the panel, the latitude was used as this would ensure that the beam of the sun is normal to the surface of the panel at solar noon. The panel is assumed to be facing due south and fixed, as fixed panels are common with utility-scale photovoltaics. With the angle of the panel determined, the beam radiation incident to the panel was calculated for all hours.

$$Solar\ Time = Standard\ Time + 4(L_{st} - L_{loc}) + E$$

$$\cos\theta_z = \cos\phi\cos\delta\cos\omega + \sin\phi\sin\delta$$

$$\cos\theta = \cos(\phi - \beta)\cos\delta\cos\omega + \sin(\phi - \beta)\sin\delta$$

$$G_b = G_{b,n}cos\theta_z$$

$$G_{b,T} = G_{b,n}cos\theta$$

$$GHI = G = G_b + G_d$$

$$G_T = G_{b,T} + G_d\left(\frac{1+\cos(\beta)}{2}\right) + \rho_g G\left(\frac{1-\cos(\beta)}{2}\right)$$

After the calculation of the environmental parameters, manufacturer specific panel details were used to calculate electricity and generation efficiency. For this project, we utilized the

PV-MLU250HC model made by Mitsubishi. This model provides a rated power of 250 W in an area of 1.66 m^2 with a module efficiency of 15.1%. We also assume that this panel has a maximum power point tracker to optimize energy production. The following equations were used for the calculation.

$$(\tau\alpha)G_T = \eta_C G_T + U_L(T_C - T_a)$$

$$(\tau\alpha)G_{NOCT} = U_L(NOCT - 20)$$

$$\eta_C = \eta_{ref}\left[1 - \beta\left(T_C - T_{C,ref}\right)\right]$$

$$\eta = \eta_{MPPT}\eta_c$$

Results

For our project, the solar farm consists of 24 million panels which has an annual energy generation of 12.3 TWh. As seen in the figure below, the month with the highest PV electricity production is May while the month with the least amount of PV energy production is in December. The efficiency ranges from 12.8 to 14.4%, decreasing in the summer months.

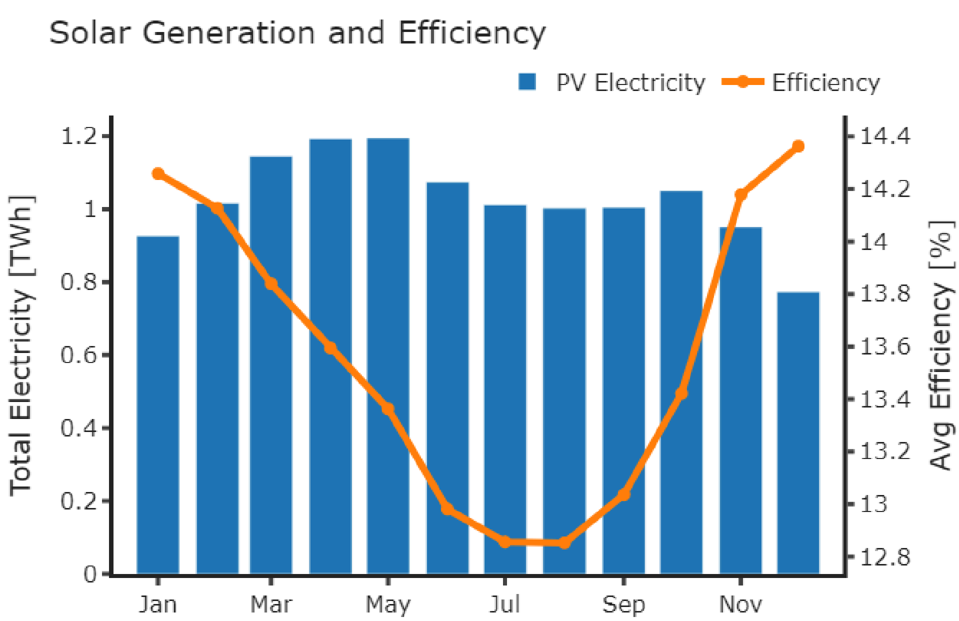

Figure 10: Monthly energy production and PV efficiency for 24 million solar panels

Sensitivity Analysis

To determine how the selection of different panels could affect performance, we did a sensitivity analysis on how variations of NOCT and the temperature coefficient of power could affect annual energy generation. The range of values was determined from the California Energy Commission's list of PV modules. However, that list did not include module efficiency as a parameter. What we found was that the temperature coefficient has a larger impact on total energy than NOCT, altering annual totals by more than 1 TWh. Both of these graphs indicate that increased temperatures will lead to a decrease in solar panel efficiency/energy production.

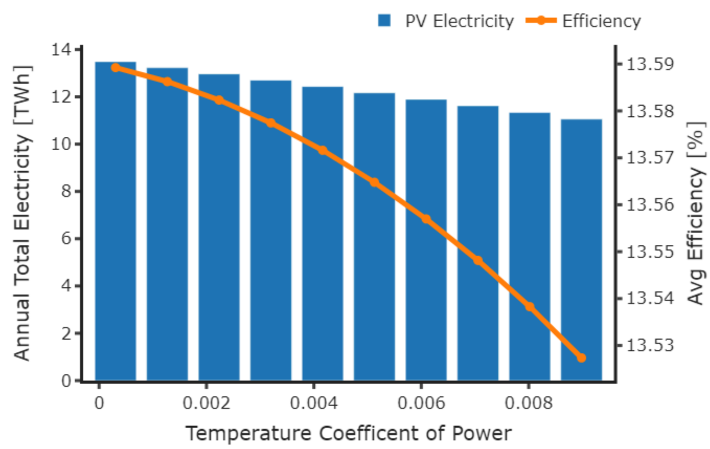

Figure 11: Impact of temperature coefficient on annual energy generation

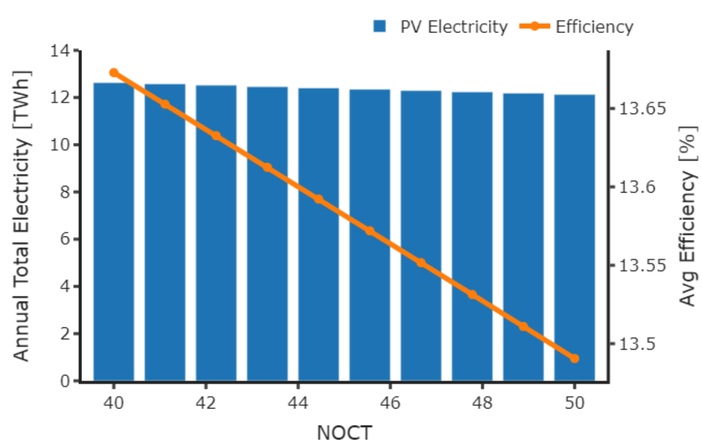

Figure 12: Impact of NOCT on annual energy generation

Siting

The area required for 24 million solar panels is 39.8 km^2, which is significantly larger than the largest solar farm in California, the Solar Star, which covers an area of 13 km^2. The site from which radiation data was taken, however, only appears to be able to fit an area of approximately 6.80 km^2, surrounded by mountainous terrain. As a result, multiple sites may be necessary in order to fit the necessary amount of solar panels.

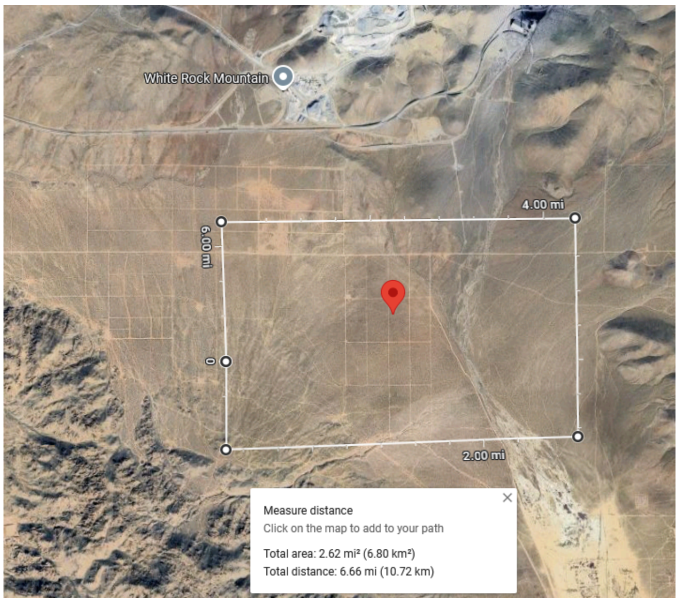

Figure 13: Possible solar farm site northeast of Los Angeles at (34.596,-117.114).

Cost

The total rated power of 24 million panels is 6 GW. In NREL's Fall 2024 Solar Industry Report, the typical costs of utility-scale solar ranged from $1-1.3/W. At $1.3 per watt, the total cost of the solar system is about $7.81 billion dollars.

Sizing Wind–Solar Solutions for the Peak Monthly Shortfall

We identified the worst-case monthly shortfall of renewable energy by analyzing the column diff_demand_minus_renewable in our monthly dataset. Specifically, we located the month with the largest gap between total demand and existing renewable generation: 2,005,339 MWh in August 2023. Addressing this peak shortfall ensures our subsequent calculations will cover the most demanding scenario—if we can meet the highest monthly deficit with additional wind and solar, all other months should be sufficiently supplied. This approach prevents underestimating renewable requirements and avoids scenarios where shorter-term deficits might leave the system reliant on non-renewable or imported energy.

For this project, three scenarios were examined to cover the August 2023 shortfall: a 50% solar / 50% wind mix, a 0% solar / 100% wind approach, and a 100% solar / 0% wind alternative. In August, each solar set (16 panels) produces roughly 0.71 MWh, while each wind turbine produces about 1006.48 MWh. Covering half of the 2,005,339 MWh gap (i.e., 1,002,670 MWh) with solar alone would therefore require 1,502,406 solar sets—equivalent to 24,038,496 panels—and the remaining half would be met by 170 wind turbines. A fully wind-based strategy (0% solar / 100% wind) would demand 340 turbines to supply the entire shortfall, whereas a completely solar strategy (100% solar / 0% wind) would require 48,076,976 panels. Figure 14 illustrates the entire spectrum from 0% solar to 100% solar for August, providing a broader view of possible wind–solar combinations.

Possible Splits to Cover the Peak Month Shortfall:

	Solar_%	Wind_%	Num_Solar_Panels	Area_Solar_m2	Num_Turbines
0	0	100	0	0.000000e+00	340
1	1	99	480784	7.979812e+05	337
2	2	98	961552	1.595936e+06	333
3	3	97	1442320	2.393891e+06	330
4	4	96	1923088	3.191845e+06	326
...	...	...	...	...	...
96	96	4	46153888	7.660392e+07	14
97	97	3	46634656	7.740187e+07	11
98	98	2	47115424	7.819982e+07	7
99	99	1	47596192	7.899778e+07	4
100	100	0	48076976	7.979576e+07	0

Figure 14: Results of Wind–Solar Splits for the Peak Monthly Shortfall

Economic analysis

An economic analysis was performed for all three scenarios using the P1–P2 method.
The equations used in this approach are outlined below, and the assumed parameters—including discount rate, inflation rate, and tax considerations—are detailed alongside.

Equations

$$P_1 = (1 - ct)PWF(N, i, d)$$

$$P_2 = D - R + (1 - D)\frac{PWF(N_1, 0, d)}{PWF(N_L, 0, m)} + [M_s(1 - ct)]PWF(N, i, d)$$

$$C_E = grid_{price} \cdot annual\ energy$$

$$C_S = first\ year\ coast$$

$$LCC_{farm} = P_2 C_S$$

$$LCC_{grid} = P_1 C_E$$

$$LCS = LCC_{grid} - LCC_{farm}$$

Values Assumed

- Wholesale electric price: \$35.90/MWh
    - Inflation: 1.5%
- Installation cost: \$15.04 B
    - Solar: \$7.81 B
    - Wind: \$7.23 B
- Total annual generation: 22.1 TWh
    - Solar: 12.3 TWh
    - Wind: 9.8 TWh
- Loan:
    - Length: 20 years
    - Interest rate: 5%
    - Down Payment ratio: 0.25
- Market discount rate: 7%
- Tax rebates: 30% from IRA

- Maintenance ratio: 4% (1.5% inflation)
- Analysis length: 25 years

Results

Table 5: P1-P2 results for different combinations of solar and wind

	50-50 Solar/Wind	All Wind	All Solar
System LCC	**$16.5 B**	**$15.5 B**	**$17.5 B**
Rated Energy	**2.21 TWh**	**2.21 TWh**	**2.21 TWh**
Grid LCC	**$1.06 B**	**$1.06 B**	**$1.06 B**
Savings	**-$15.4 B**	**-$14.4 B**	**-$16.4 B**
LCOE:	**$298.30/MWh**	**$279.30/MWh**	**$316.85/MWh**

As seen in Table 5, the all wind scenario appears to offer the best cost for the same amount of rated power produced. The savings appear to get more negative with increased analysis length, so it is unclear what the payback period is. There is a possibility that the cost of maintenance and the fairly low wholesale price of $35 compared to LCOE of ~$300 may be making it more difficult to pay off the system. To that end, we also performed an analysis to determine the overall system cost where savings would even be possible and the results are shown below.

Table 6: P1-P2 results for breakeven system LCC

	Break-even farm cost
System LCC	**$1.06 B**
Rated Energy	**2.21 TWh**
Grid LCC	**$1.06 B**
Savings	**$336k**
LCOE:	**$19.12/MWh**

As shown above, the system cost would need to decrease from $16.5 B to $1.06 B in order to achieve savings. At this system cost the payback period would be 24 years and adjusting the analysis length shows that savings turn positive with increased number of years from -$60B with an analysis length of 20 years to $47 B with an analysis length of 30 years.

Conclusion

This study demonstrates that, while Los Angeles can feasibly meet its worst-case monthly shortfall in August through expanded wind and solar resources, the overall system costs remain substantial. Among the scenarios examined, an all-wind configuration appears the most economically favorable, though both purely solar and mixed solar–wind solutions also achieve net-zero outcomes.

One major limitation of this analysis is that it does not consider energy storage. The calculations assume solar and wind resources can directly cover demand at all times, neglecting the reality that supply–demand mismatches occur over short or extended periods. A logical next step would be to integrate storage technologies into the model. Doing so would allow for a more accurate assessment of optimal storage capacity, cost, and operational strategies for charging and discharging.

Another limitation involves the exclusion of transmission constraints. The present study assumes that large-scale solar and wind installations can feed power into the grid without encountering congestion or line losses. Future work should incorporate grid capacity assessments and potential costs for transmission line expansions or upgrades. Evaluating these infrastructure requirements is key to determining the feasibility of siting decisions and ensuring reliable power delivery.

A further improvement would be to move beyond a simplistic approach to wind–solar combinations and consider other potential renewables, such as hydro and geothermal. Exploring a wider range of renewable sources could provide a more robust and flexible energy mix. Meanwhile, on the environmental front, expanded research should incorporate air quality impacts, along with associated public health and economic benefits—such as reductions in asthma and related healthcare costs—to fully capture the value of transitioning away from fossil-fuel-based generation.

Code

The code for this project is publically available at:
https://github.com/Sustainable-Building-Energy/Sustainable

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